
MSML 605

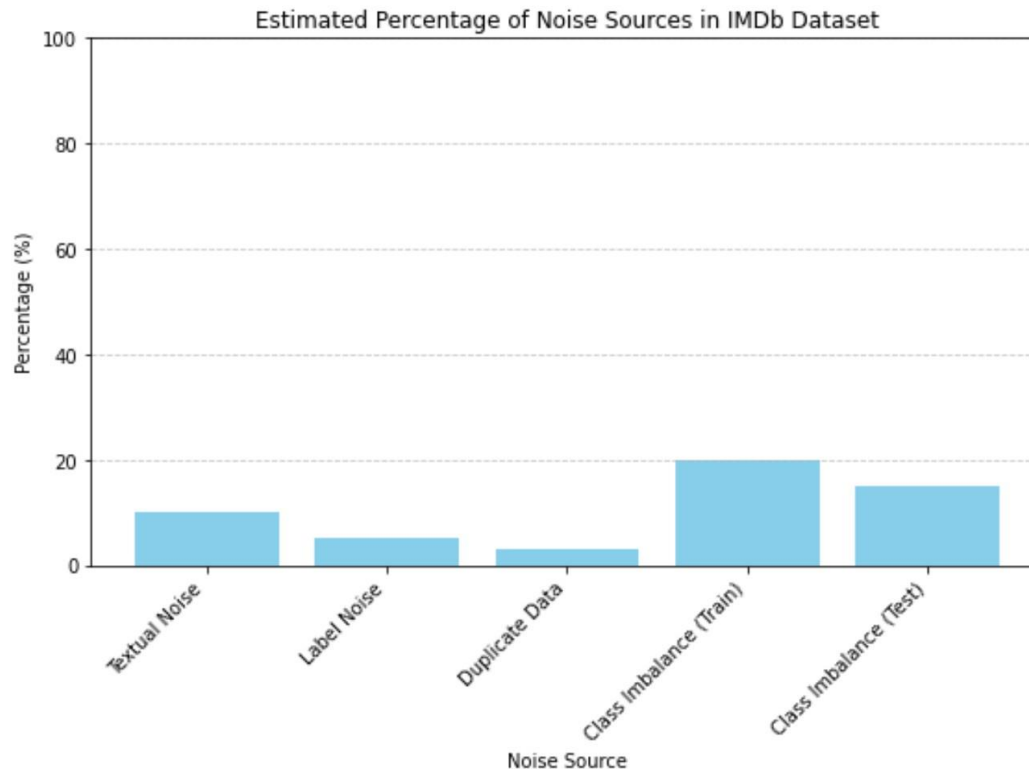
Error Analysis

Introduction

- There are different types of noise levels in a dataset.
 - We want to be able to prioritize what to work on.
 - For example, text dataset
 - Textual noise
 - Label noise
 - Duplicate data
 - Class imbalance
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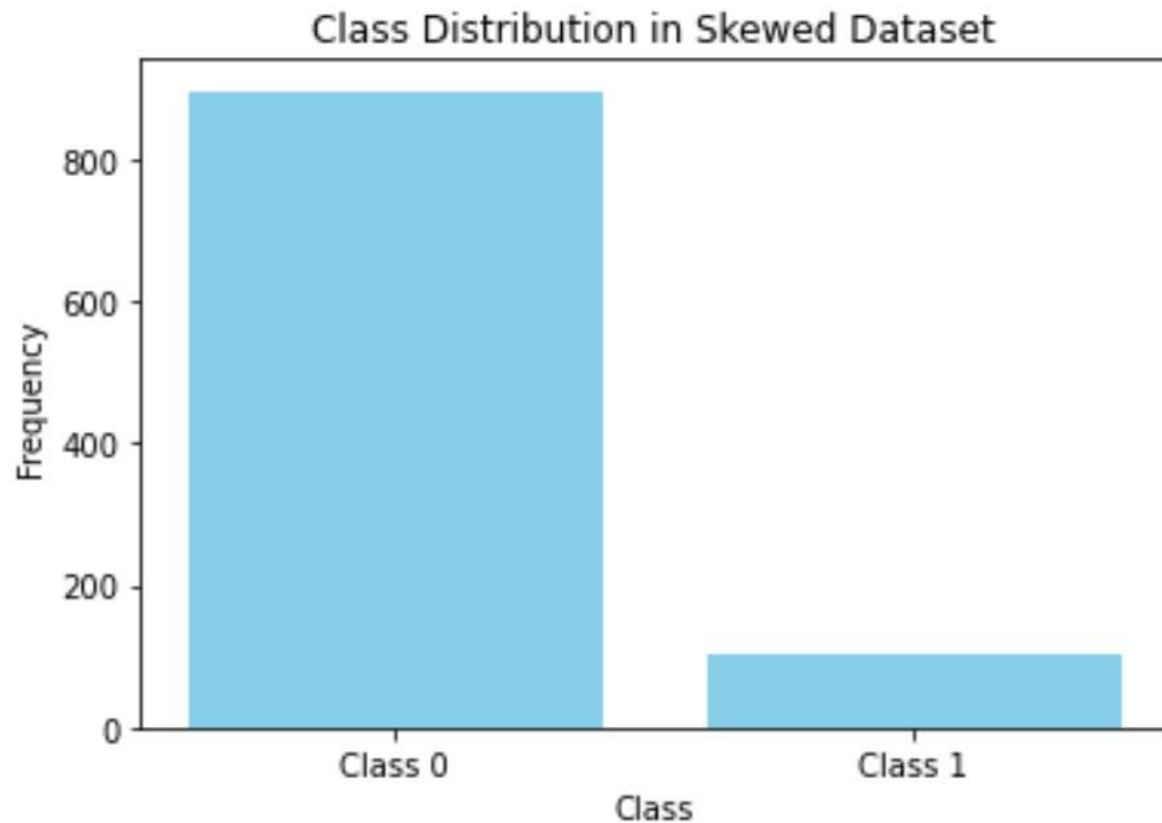
Prioritize what to be done

Percentage of Textual Noise: 10.00%
Percentage of Label Noise: 5.00%
Percentage of Duplicate Data: 3.00%
Percentage of Class Imbalance in Train set: 20.00%
Percentage of Class Imbalance in Test set: 15.00%



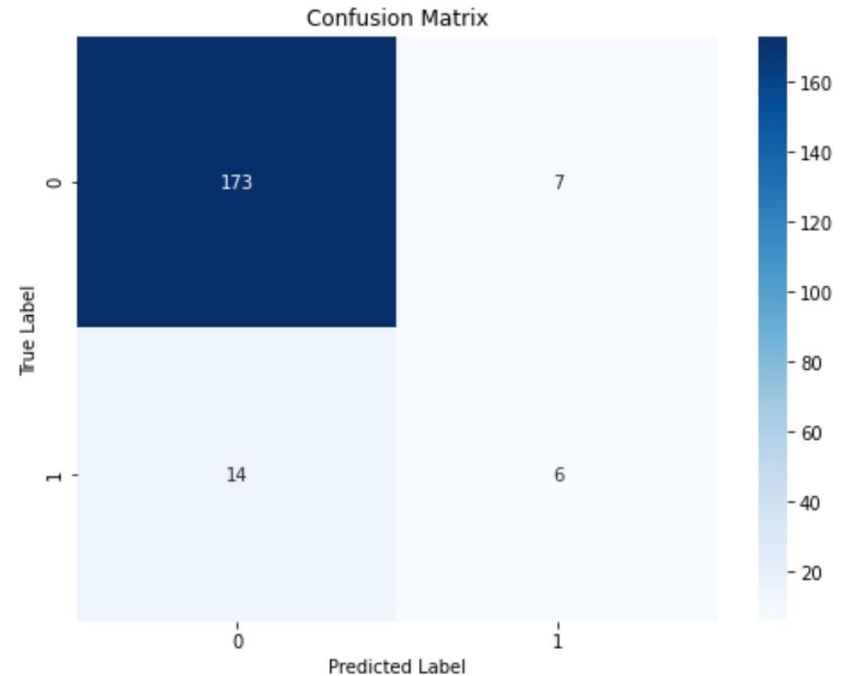
Skewed Data

Percentage of Class Imbalance: 89.70%



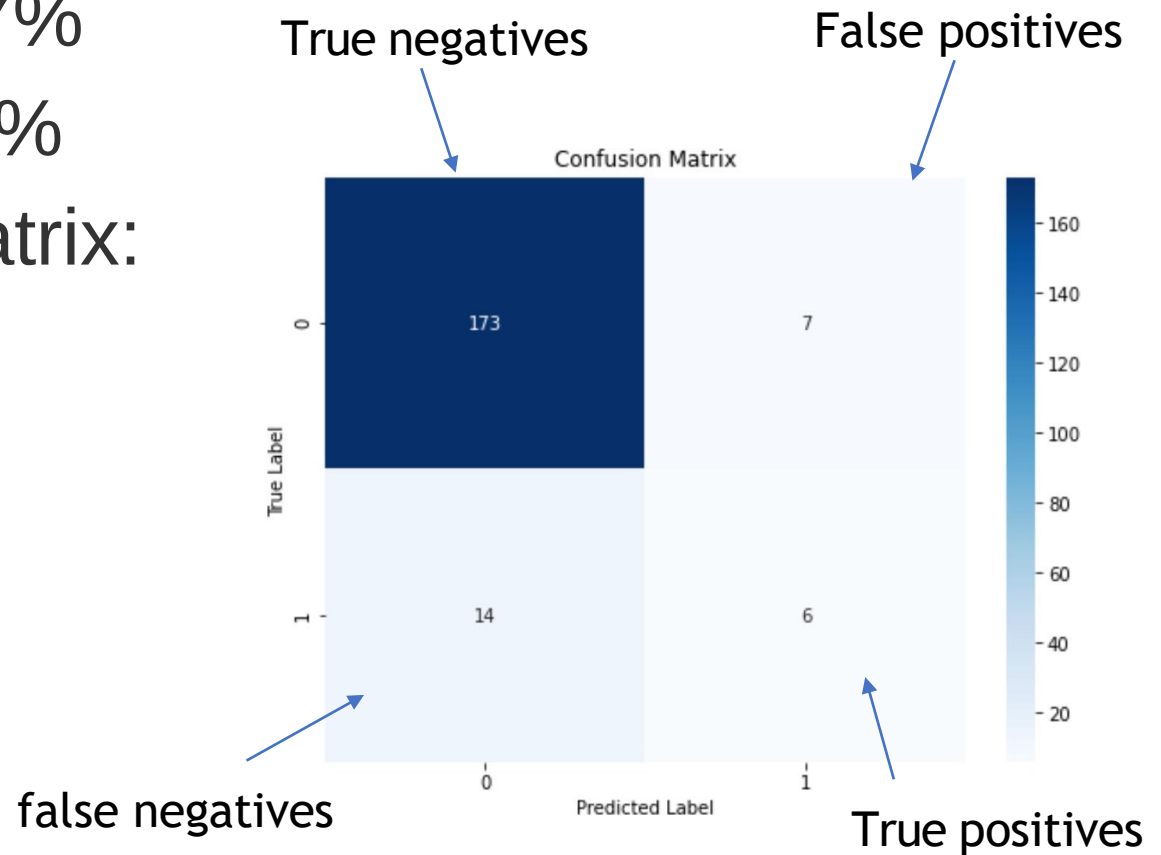
Skewed Data

- Class:
 - $y = 0$, 89.7%
 - $Y = 1$, 10.3 %
- Confusion matrix:

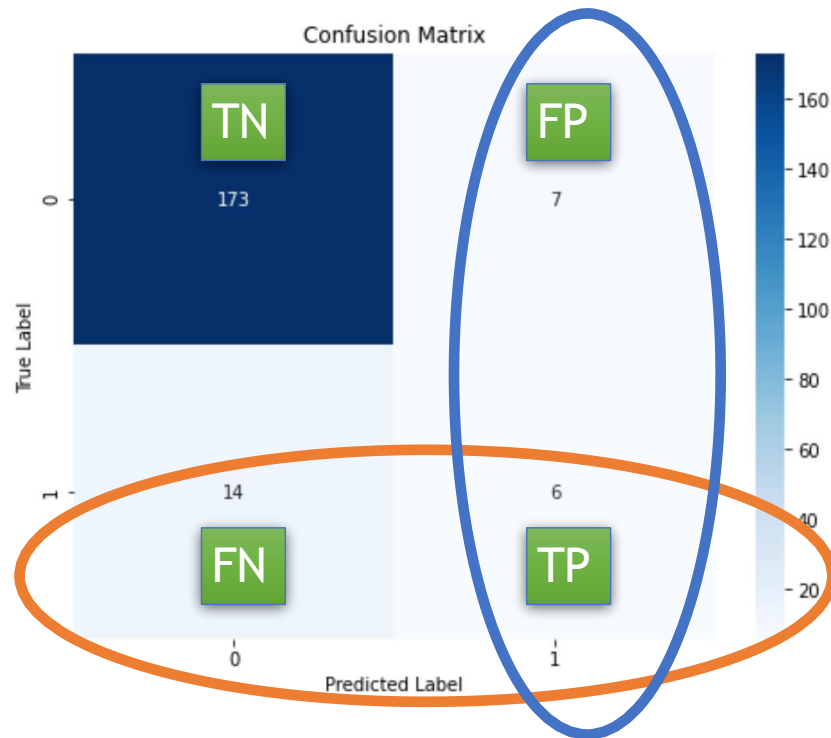


Skewed Data

- Class:
 - $y = 0$, 89.7%
 - $y = 1$, 10.3 %
- Confusion matrix:



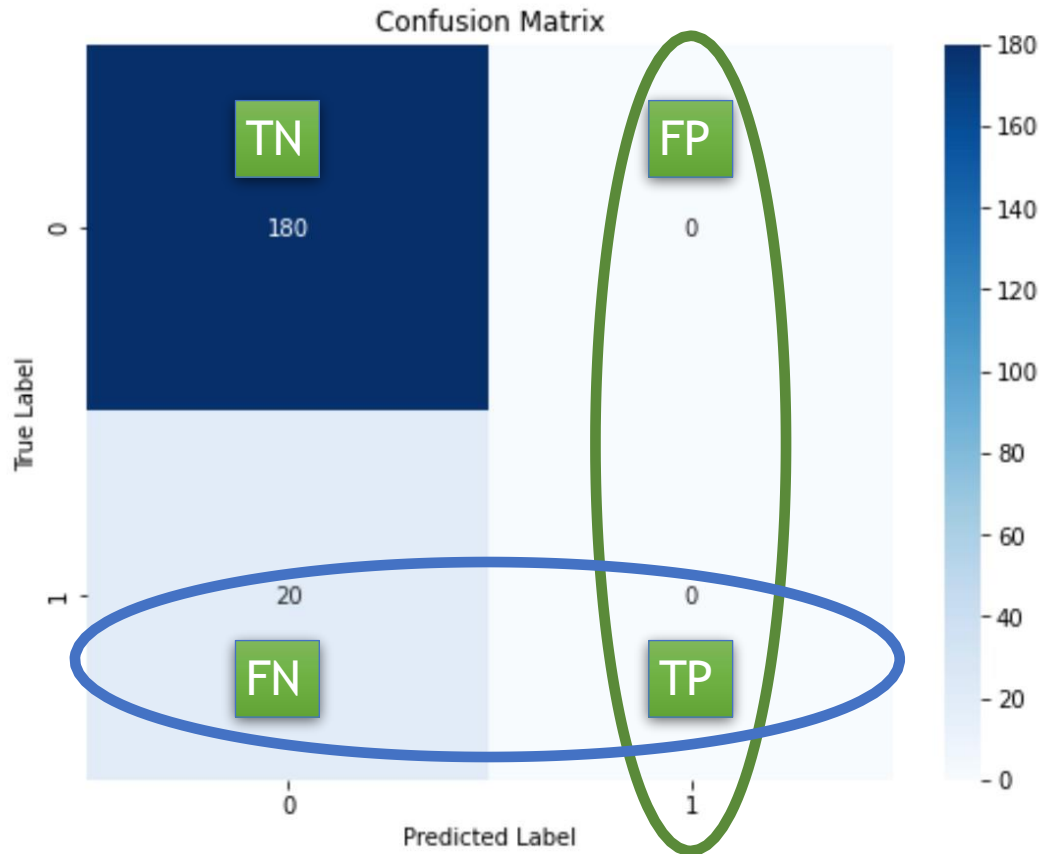
Confusion matrix



$$Precision = \frac{TP}{TP + FP} = \frac{6}{6 + 7} = 46.15 \%$$

$$Recall = \frac{TP}{TP + FN} = \frac{6}{6 + 14} = 30 \%$$

When prediction is 0



$$Precision = \frac{TP}{TP + FP} = \frac{0}{0 + 0} = 0\%$$

$$Recall = \frac{TP}{TP + FN} = \frac{0}{0 + 20} = 0\%$$

F1 Score

	Precision	Recall	Accuracy	F1- score
Model 1 (logistic Regression)	46.15 %	30 %	89.5 %	0.36
Model 2 (print 0)	0 %	0 %	90 %	0.0

$$F_1 = \frac{2}{\frac{1}{P} + \frac{1}{R}}$$

Multi-class classification

	Class 0	Class 1	Class 2	Class 3
Class 0	116	8	0	1
Class 1	6	10	0	1
Class 2	2	0	24	0
Class 3	11	0	0	21

F1 Score: 0.8532455544455546

Multi-class classification

	Precision	Recall	F1-score
Class 0	0.859	0.928	0.892
Class 1	0.555	0.588	0.571
Class 2	1.0	0.923	0.960
Class 3	0.913	0.656	0.763

$$precision_i = \frac{TP_i}{TP_i + FP_i}$$

$$recall_i = \frac{TP_i}{TP_i + FN_i}$$

Check performance

- Brainstorm
 - Performance on a subset of data.
 - How does it perform on rare observations?
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